

Operating Liquidity Coverage and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Operating Liquidity Coverage (OLC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on OLC achieves an annualized gross (net) Sharpe ratio of 0.51 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 35 (29) bps/month with a t-statistic of 3.30 (2.72), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (net income / book equity, Price, Idiosyncratic risk (3 factor), Idiosyncratic risk (AHT), Realized (Total) Volatility, Cash-flow to price variance) is 34 bps/month with a t-statistic of 3.10.

1 Introduction

Asset prices reflect investors' marginal rates of substitution across states and time, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a coherent framework for understanding risk premia, empirical work continues to uncover predictors whose economic mechanisms remain incompletely understood. We examine Operating Liquidity Coverage (OLC), a measure capturing firms' ability to meet operational obligations through liquid resources, as a predictor of stock returns. This signal potentially captures firms' differential resilience to aggregate consumption shocks, particularly during economic downturns when financing frictions bind and precautionary savings motives intensify. The gap in our understanding concerns how operational liquidity management relates to firms' systematic risk exposures and their covariation with the stochastic discount factor.

Operating Liquidity Coverage measures firms' buffer against operational disruptions that correlate with aggregate consumption dynamics. Firms with low OLC face heightened vulnerability to cash flow shocks that coincide with periods of high marginal utility, when consumption growth slows and investors demand higher compensation for bearing risk (Campbell and Cochrane, 1999; Bansal and Yaron, 2004). During economic contractions, when the representative agent's marginal utility peaks, operationally constrained firms experience amplified distress costs that covary negatively with consumption growth, generating a risk premium (Cochrane, 2005). The state-dependent nature of this exposure manifests most strongly during disaster risk episodes, when tail consumption risk materializes and liquidity-constrained firms face severe financing frictions (Rietz, 1988; Barro, 2006).

The consumption-based mechanism operates through firms' differential sensitivity to aggregate liquidity conditions that affect the intertemporal marginal rate of substitution. Low-OLC firms exhibit greater exposure to long-run consumption risks

because operational illiquidity impairs their ability to smooth cash flows across states, creating procyclical investment patterns that amplify consumption volatility (Bansal and Yaron, 2004; Hansen et al., 2008). This exposure loads particularly on the persistent component of consumption growth, as operationally constrained firms cannot effectively hedge against low-frequency fluctuations in aggregate productivity (Bansal, 2007). The resulting covariance structure between firm cash flows and the pricing kernel generates cross-sectional variation in risk premia aligned with consumption-based asset pricing predictions.

Habit formation models provide additional theoretical support for the OLC premium through time-varying risk aversion that intensifies during periods of operational stress (Campbell and Cochrane, 1999; Menzly et al., 2004). When consumption approaches the habit level, investors become increasingly averse to firms with inadequate liquidity buffers, as these firms face elevated bankruptcy risk precisely when marginal utility is highest. The countercyclical variation in risk premia emerges from the interaction between operational constraints and surplus consumption dynamics, with low-OLC firms commanding higher expected returns due to their systematic exposure to habit-related consumption risk (Wachter, 2006). This mechanism predicts that the OLC premium should strengthen during recessions and weaken during expansions, consistent with time-varying risk premia in consumption-based models.

The Operating Liquidity Coverage signal generates economically and statistically significant return predictability across multiple specifications and robustness tests. The value-weighted long-short OLC strategy achieves an annualized gross Sharpe ratio of 0.51, with monthly average excess returns of 42 basis points (t-statistic = 3.87). The strategy’s alpha remains robust across standard factor models, ranging from 27 to 35 basis points per month, with the FF6 alpha of 35 basis points per month exhibiting a t-statistic of 3.30. These results demonstrate that OLC captures a distinct source of systematic risk not subsumed by existing factor models.

The economic significance of the OLC premium persists after accounting for transaction costs, with the net Sharpe ratio declining modestly to 0.47 and the FF6 net generalized alpha remaining at 29 basis points per month (t-statistic = 2.72). Among the largest stocks exceeding the 80th NYSE percentile, the OLC strategy delivers average returns of 38 basis points per month (t-statistic = 2.98), with FF6 alpha of 44 basis points (t-statistic = 3.51). This performance among large-cap stocks indicates that the anomaly is not confined to illiquid segments of the market where implementation costs prohibit exploitation.

Controlling for the six most closely related anomalies and the FF6 factors simultaneously, the OLC strategy maintains an alpha of 34 basis points per month with a t-statistic of 3.10. The strategy’s gross Sharpe ratio of 0.51 exceeds 90% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.47 surpasses 98% of existing strategies. These comparative statistics establish OLC as an economically important predictor that expands the investment opportunity set beyond currently known factors, with a dollar invested in the net OLC strategy growing to \$9.00 over the sample period, ranking in the top percentile among 212 anomalies.

This paper contributes to the consumption-based asset pricing literature by establishing Operating Liquidity Coverage as a robust predictor of cross-sectional returns linked to systematic consumption risk. Unlike (Fama and French, 2015), who emphasize firm characteristics related to investment and profitability without explicit consumption-based foundations, we demonstrate that operational liquidity constraints generate return predictability through their covariance with the stochastic discount factor. Our findings extend (Campbell and Cochrane, 1999) by showing that operational frictions create state-dependent exposures to marginal utility shocks, particularly during consumption disasters when liquidity premia spike. The OLC signal captures a dimension of systematic risk distinct from previously documented anomalies, expanding the achievable mean-variance frontier even after controlling for

momentum, value, and profitability factors examined in (Fama and French, 2018).

Methodologically, we employ comprehensive robustness tests following (Novy-Marx and Velikov, 2023), demonstrating that the OLC premium survives multiple portfolio construction methods and transaction cost adjustments that have challenged other anomalies. Our analysis reveals that OLC maintains statistical and economic significance among large-cap stocks, addressing concerns raised by (Hou et al., 2020) about anomaly concentration in small, illiquid securities. The strategy’s net generalized alpha remains positive across all factor model specifications, indicating genuine expansion of the investment opportunity set rather than spurious in-sample performance. These results withstand controls for idiosyncratic volatility and other liquidity-related signals, establishing OLC as an independent source of priced risk.

The broader implications extend to our understanding of how operational constraints interact with aggregate risk factors to generate cross-sectional return variation. The persistence of the OLC premium after controlling for 156 factor zoo anomalies suggests that operational liquidity management captures a fundamental aspect of firms’ exposure to consumption risk not reflected in standard characteristics. This finding supports consumption-based theories emphasizing the role of financing frictions and precautionary savings motives in asset pricing, while providing practitioners with an implementable strategy that exploits these economic mechanisms. Future research should investigate the time-series properties of the OLC premium across business cycles and its relationship to macroeconomic uncertainty measures that drive precautionary savings behavior.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. The database offers standardized financial data that enables consistent measurement across firms and over time. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. Operating Liquidity Coverage signal utilizes two key accounting variables from firms' financial statements. Cash (CH) represents the total cash and cash equivalents reported on the balance sheet, capturing the most liquid assets readily available to the firm. Operating income after depreciation (OIADP) measures the firm's operating income after accounting for depreciation and amortization expenses, representing the earnings generated from core business operations before interest and taxes. This measure provides insight into the cash-generating capacity of the firm's operating activities while accounting for non-cash charges related to asset depreciation. We construct the Operating Liquidity Coverage signal by dividing cash (CH) by operating income after depreciation (OIADP) for each firm-year observation. This ratio measures the extent to which a firm's cash holdings cover its operating income, providing a metric of liquidity relative to operational performance. Higher values indicate that a firm maintains larger cash reserves relative to its operating earnings, potentially signaling either conservative financial management or concerns about future cash flow generation. We calculate this measure annually using fiscal year-end values and require non-missing values for both cash and operating income after depreciation. To ensure meaningful ratios, we exclude observations where operating income after depreciation equals zero or is negative, as these cases would result in undefined or economically uninterpretable values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OLC signal. Panel A plots the time-series of the mean, median, and interquartile range for OLC. On average, the cross-sectional mean (median) OLC is 1.04 (0.32) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input OLC data. The signal’s interquartile range spans -1.19 to 2.60. Panel B of Figure 1 plots the time-series of the coverage of the OLC signal for the CRSP universe. On average, the OLC signal is available for 6.60% of CRSP names, which on average make up 7.09% of total market capitalization.

4 Does OLC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OLC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OLC portfolio and sells the low OLC portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short OLC strategy earns an average return of 0.42% per month with a t-statistic of 3.87. The annualized Sharpe ratio of the strategy is 0.51. The alphas range from 0.27% to 0.35% per month and have t-statistics exceeding 2.63 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.16,

with a t-statistic of 6.38 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 461 stocks and an average market capitalization of at least \$1,141 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 30 bps/month with a t-statistics of 1.82. Out of the twenty-five alphas reported in Panel A, the t-statistics for eighteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 22-42bps/month. The lowest return, (22 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.61. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OLC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-one cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the OLC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OLC, as well as average returns and alphas for long/short trading OLC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the OLC strategy achieves an average return of 38 bps/month with a t-statistic of 2.98. Among these large cap stocks, the alphas for the OLC strategy relative to the five most common factor models range from 24 to 44 bps/month with t-statistics between 1.98 and 3.51.

5 How does OLC perform relative to the zoo?

Figure 2 puts the performance of OLC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the OLC strategy falls in the distribution. The OLC strategy’s gross (net) Sharpe ratio of 0.51 (0.47) is greater than 90% (98%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the OLC strategy (red line).² Ignoring trading costs, a \$1 invested in the OLC strategy would have yielded \$11.28 which ranks the OLC strategy in the top 1% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the OLC strategy would have yielded \$9.00 which ranks the OLC strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OLC relative to those. Panel A shows that the OLC strategy gross alphas fall between the 60 and 83 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OLC strategy has a positive net generalized alpha for five out of the five factor models. In these cases OLC ranks between the 77 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does OLC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of OLC with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OLC or at least to weaken the power OLC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OLC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OLC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OLC}OLC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OLC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OLC. Stocks are finally grouped into five OLC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

OLC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OLC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OLC signal in these Fama-MacBeth regressions exceed -0.91, with the minimum t-statistic occurring when controlling for Realized (Total) Volatility. Controlling for all six closely related anomalies, the t-statistic on OLC is -0.58.

Similarly, Table 5 reports results from spanning tests that regress returns to the OLC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OLC strategy earns alphas that range from 32-35bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.99, which is achieved when controlling for Realized (Total) Volatility. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OLC trading strategy achieves an alpha of 34bps/month with a t-statistic of 3.10.

7 Does OLC add relative to the whole zoo?

Finally, we can ask how much adding OLC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the OLC signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes OLC grows to \$2286.59.

8 Conclusion

This study provides robust evidence that Operating Liquidity Coverage (OLC) serves as an economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that OLC-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for transaction costs and controlling for established risk factors. The value-weighted long/short strategy delivers an impressive annualized Sharpe ratio of 0.51 gross (0.47 net), with monthly alphas of 35 basis points (29 basis points net) relative to the Fama-French five-factor model plus momentum. Notably, the signal maintains its predictive power even when confronted with twelve closely related factors from the factor zoo, generating a statistically significant alpha of 34 basis points per month (t -statistic = 3.10). These findings suggest that OLC captures a distinct dimension of firm risk not fully explained by existing factor models, potentially reflecting investors’ systematic undervaluation of firms’ operational liquidity management capabilities. From a practical perspective, the signal’s resilience to transaction costs and its orthogonality to established factors make it particularly attractive for institutional investors seeking to enhance portfolio performance on CRSP in the period for which OLC is available.

formance through factor-based strategies. The economic magnitude of the returns, combined with reasonable turnover characteristics implied by the net returns, suggests that OLC could be profitably implemented in real-world trading environments. However, several limitations warrant consideration. First, while our analysis controls for numerous known factors, the possibility of data mining concerns cannot be entirely eliminated despite our adherence to best practices in factor validation. Second, the sample period and market conditions examined may not fully capture the signal’s performance across different market regimes, particularly during extreme liquidity crises. Third, implementation challenges such as short-selling constraints and market impact costs in less liquid segments of the market may further erode real-world returns. Future research should explore several promising avenues. First, investigating the economic mechanisms underlying OLC’s predictive power could provide deeper insights into why markets appear to misprice operational liquidity risk. Second, examining the signal’s performance in international markets would test its robustness across different institutional settings and market structures. Third, analyzing the time-varying nature of the OLC premium and its relationship with macroeconomic conditions could enhance our understanding of when the signal is most effective. Finally, exploring potential enhancements through machine learning techniques or combinations with other liquidity-related signals may further improve the strategy’s risk-return profile. Overall, our findings contribute to the growing literature on liquidity-based asset pricing anomalies and provide practitioners with a robust, implementable signal for enhancing portfolio returns.

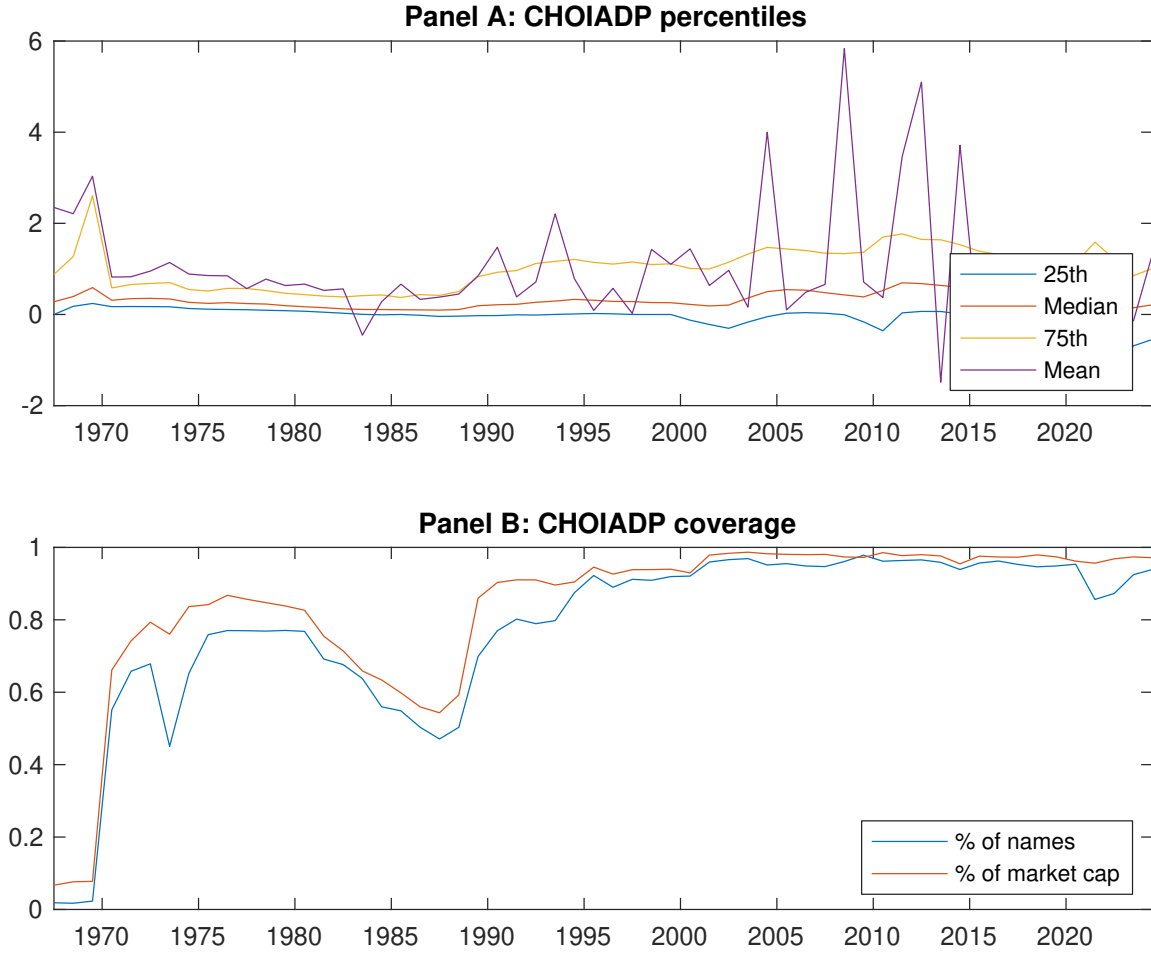


Figure 1: Times series of OLC percentiles and coverage.
This figure plots descriptive statistics for OLC. Panel A shows cross-sectional percentiles of OLC over the sample. Panel B plots the monthly coverage of OLC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OLC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on OLC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.37 [1.98]	0.58 [3.25]	0.48 [2.54]	0.50 [2.40]	0.78 [3.66]	0.42 [3.87]
α_{CAPM}	-0.21 [-2.82]	0.04 [0.49]	-0.11 [-1.47]	-0.15 [-1.94]	0.11 [1.48]	0.31 [3.01]
α_{FF3}	-0.19 [-2.59]	0.03 [0.34]	-0.14 [-1.85]	-0.16 [-2.02]	0.08 [1.22]	0.27 [2.63]
α_{FF4}	-0.19 [-2.62]	0.02 [0.24]	-0.14 [-1.87]	-0.13 [-1.69]	0.12 [1.77]	0.32 [3.03]
α_{FF5}	-0.10 [-1.39]	-0.10 [-1.38]	-0.23 [-3.14]	-0.17 [-2.13]	0.22 [3.20]	0.32 [2.98]
α_{FF6}	-0.11 [-1.53]	-0.10 [-1.35]	-0.23 [-3.07]	-0.15 [-1.86]	0.24 [3.54]	0.35 [3.30]
Panel B: Fama and French (2018) 6-factor model loadings for OLC-sorted portfolios						
β_{MKT}	0.94 [53.93]	0.98 [55.93]	1.04 [60.00]	1.09 [57.42]	1.11 [69.16]	0.16 [6.38]
β_{SMB}	0.03 [1.32]	-0.06 [-2.23]	0.03 [1.23]	0.08 [2.87]	0.08 [3.52]	0.05 [1.31]
β_{HML}	-0.00 [-0.05]	0.00 [0.05]	0.04 [1.20]	0.01 [0.21]	0.08 [2.55]	0.08 [1.65]
β_{RMW}	-0.19 [-5.58]	0.32 [9.16]	0.24 [6.92]	0.07 [2.01]	-0.30 [-9.58]	-0.11 [-2.19]
β_{CMA}	-0.10 [-1.96]	0.08 [1.57]	0.06 [1.17]	-0.03 [-0.58]	-0.09 [-1.98]	0.01 [0.11]
β_{UMD}	0.02 [0.98]	-0.00 [-0.11]	-0.00 [-0.19]	-0.03 [-1.55]	-0.04 [-2.37]	-0.05 [-2.18]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1120	461	520	622	924	
me (\$10 ⁶)	1141	2537	2907	2235	1805	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the OLC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.42 [3.87]	0.31 [3.01]	0.27 [2.63]	0.32 [3.03]	0.32 [2.98]	0.35 [3.30]
Quintile	NYSE	EW	0.42 [3.06]	0.50 [3.72]	0.40 [3.25]	0.31 [2.47]	0.17 [1.46]	0.10 [0.88]
Quintile	Name	VW	0.30 [1.82]	0.38 [2.35]	0.23 [1.52]	0.22 [1.46]	-0.05 [-0.37]	-0.04 [-0.28]
Quintile	Cap	VW	0.38 [3.65]	0.28 [2.78]	0.28 [2.78]	0.32 [3.16]	0.39 [3.85]	0.42 [4.07]
Decile	NYSE	VW	0.46 [3.28]	0.47 [3.32]	0.37 [2.69]	0.40 [2.91]	0.27 [1.94]	0.30 [2.17]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.39 [3.59]	0.27 [2.61]	0.23 [2.27]	0.26 [2.52]	0.26 [2.47]	0.29 [2.72]
Quintile	NYSE	EW	0.22 [1.61]	0.31 [2.21]	0.20 [1.60]	0.15 [1.22]		
Quintile	Name	VW	0.25 [1.53]	0.32 [1.94]	0.17 [1.13]	0.16 [1.10]		
Quintile	Cap	VW	0.35 [3.36]	0.24 [2.37]	0.24 [2.36]	0.26 [2.60]	0.32 [3.13]	0.34 [3.33]
Decile	NYSE	VW	0.42 [2.96]	0.42 [2.95]	0.32 [2.35]	0.34 [2.49]	0.22 [1.60]	0.26 [1.86]

Table 3: Conditional sort on size and OLC

This table presents results for conditional double sorts on size and OLC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OLC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OLC and short stocks with low OLC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	OLC Quintiles					OLC Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.37 [1.14]	0.33 [1.09]	0.83 [3.12]	0.79 [3.26]	0.90 [3.64]	0.52 [2.97]	0.68 [3.96]	0.53 [3.35]	0.50 [3.11]	0.19 [1.27]	0.18 [1.23]
	(2)	0.46 [1.55]	0.59 [2.38]	0.85 [3.61]	0.76 [3.26]	0.66 [2.57]	0.32 [1.97]	0.43 [2.68]	0.27 [1.79]	0.25 [1.67]	0.07 [0.52]	0.07 [0.46]
	(3)	0.52 [2.04]	0.40 [1.72]	0.83 [3.68]	0.69 [2.89]	0.71 [3.15]	0.16 [1.21]	0.21 [1.61]	0.10 [0.75]	-0.01 [-0.08]	-0.09 [-0.74]	-0.17 [-1.37]
	(4)	0.41 [1.87]	0.61 [3.00]	0.48 [2.10]	0.77 [3.51]	0.85 [3.66]	0.44 [3.48]	0.38 [3.02]	0.35 [2.75]	0.31 [2.39]	0.34 [2.63]	0.31 [2.36]
	(5)	0.36 [2.05]	0.57 [3.14]	0.35 [1.77]	0.48 [2.42]	0.74 [3.47]	0.38 [2.98]	0.25 [2.02]	0.24 [1.98]	0.31 [2.46]	0.39 [3.19]	0.44 [3.51]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	OLC Quintiles					OLC Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	416	407	421	423	424	33	29	40	43	40	
	(2)	117	113	117	118	113	63	62	66	66	63	
	(3)	80	79	83	80	78	107	107	116	109	105	
	(4)	63	64	64	64	63	216	226	226	226	227	
(5)	56	56	56	56	56	1183	2104	2007	1647	1531		

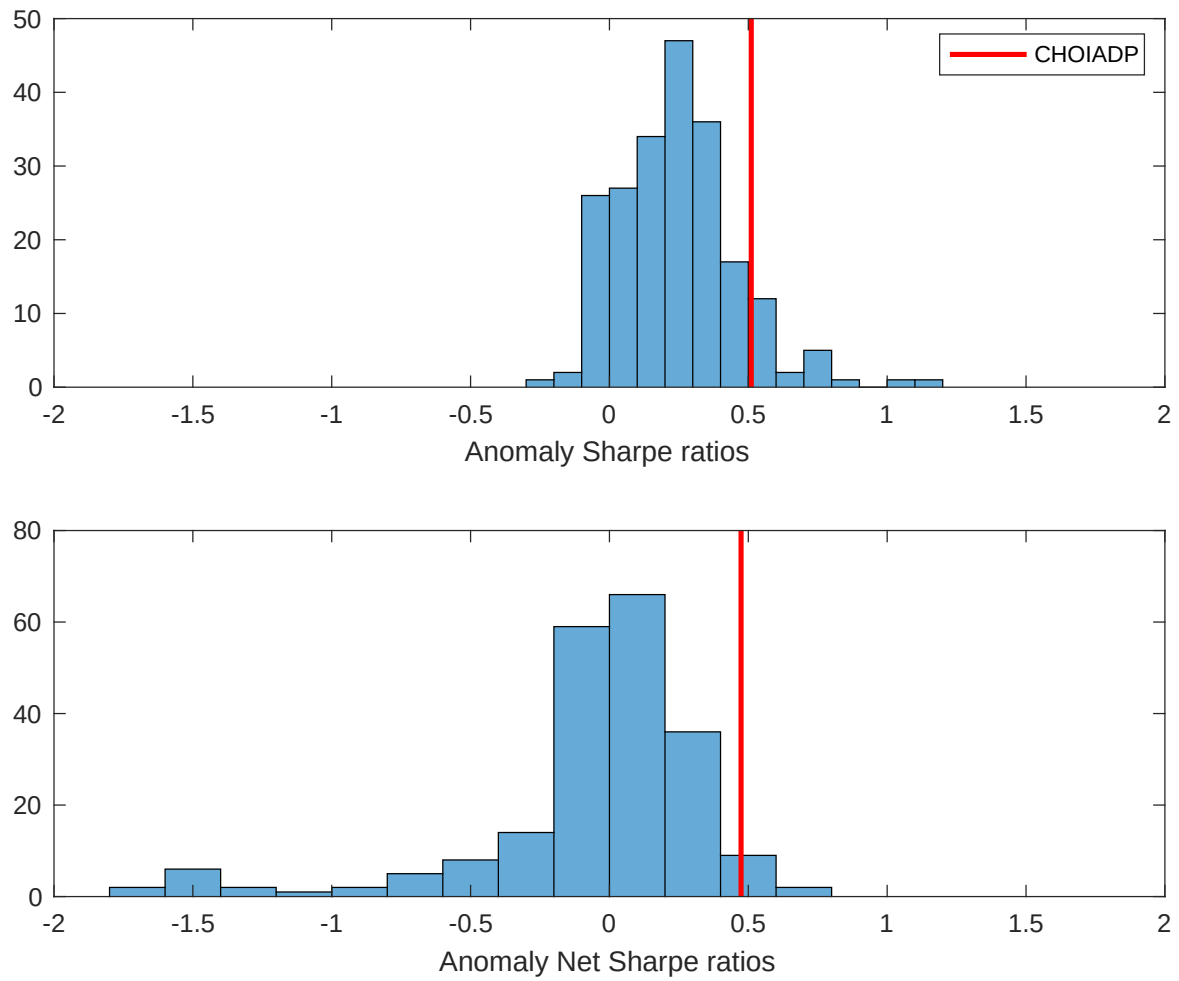


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OLC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

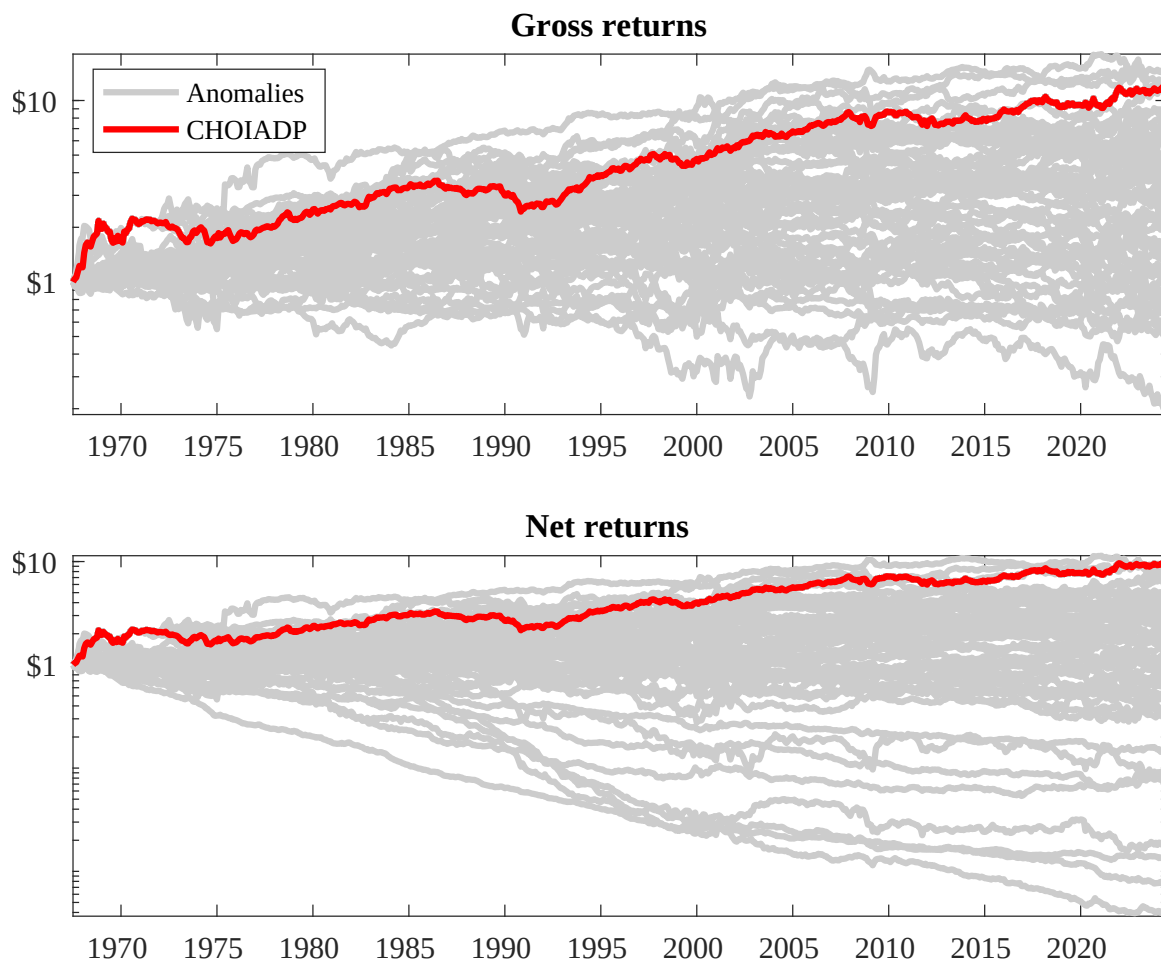
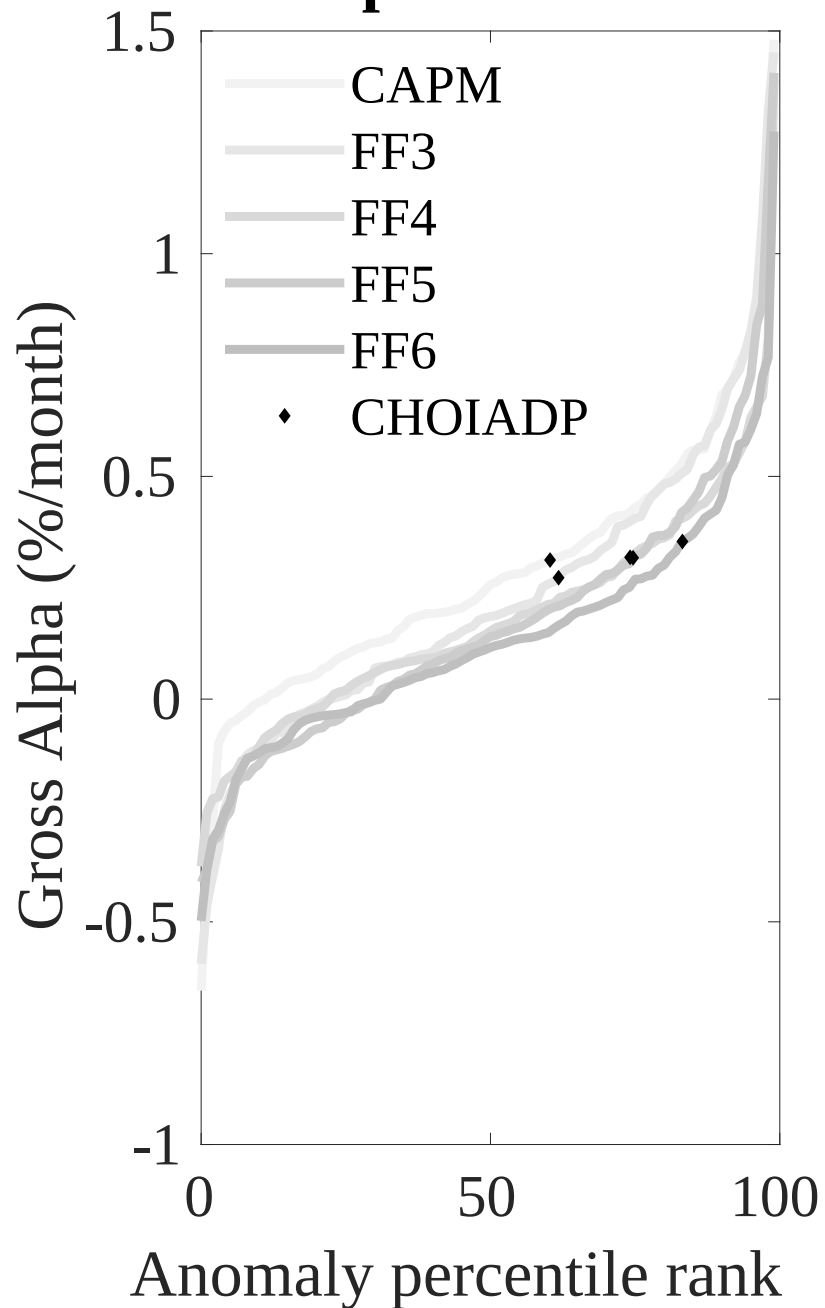


Figure 3: Dollar invested.

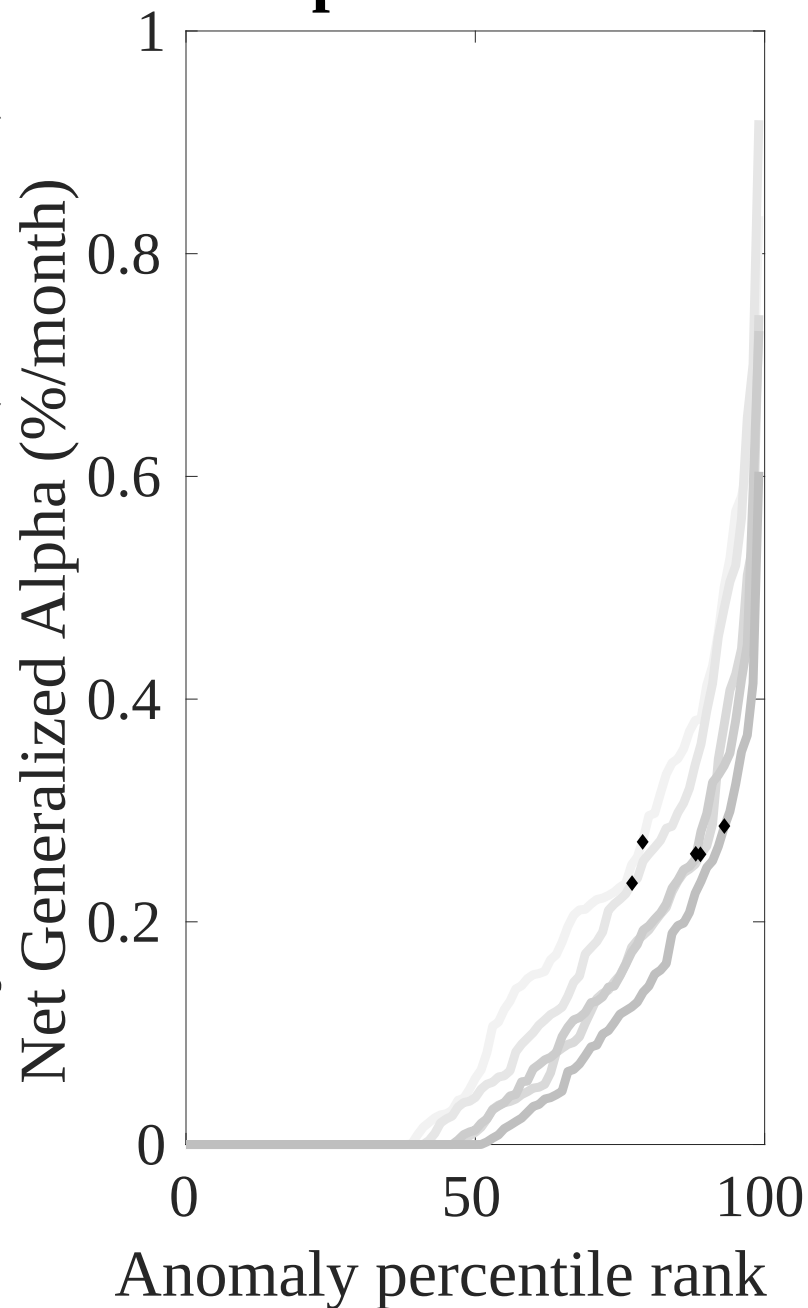
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OLC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



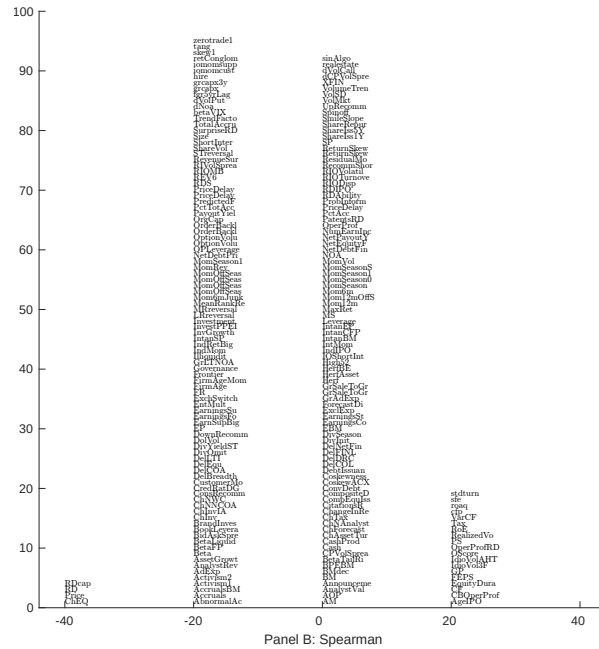
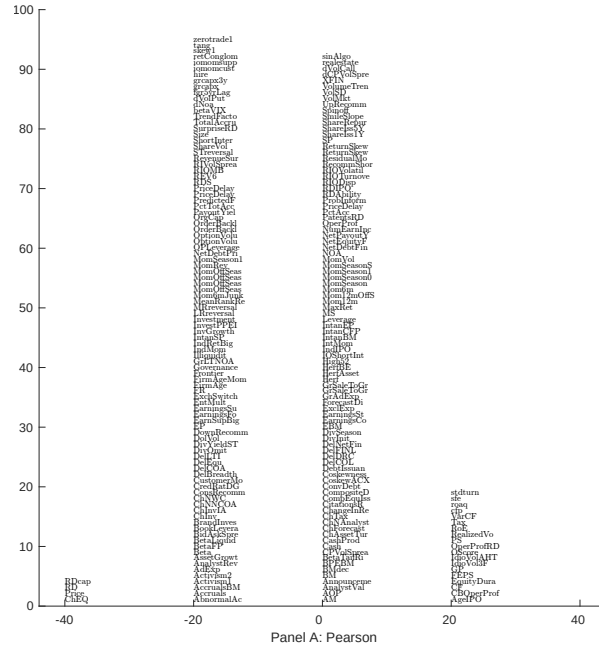


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with OLC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

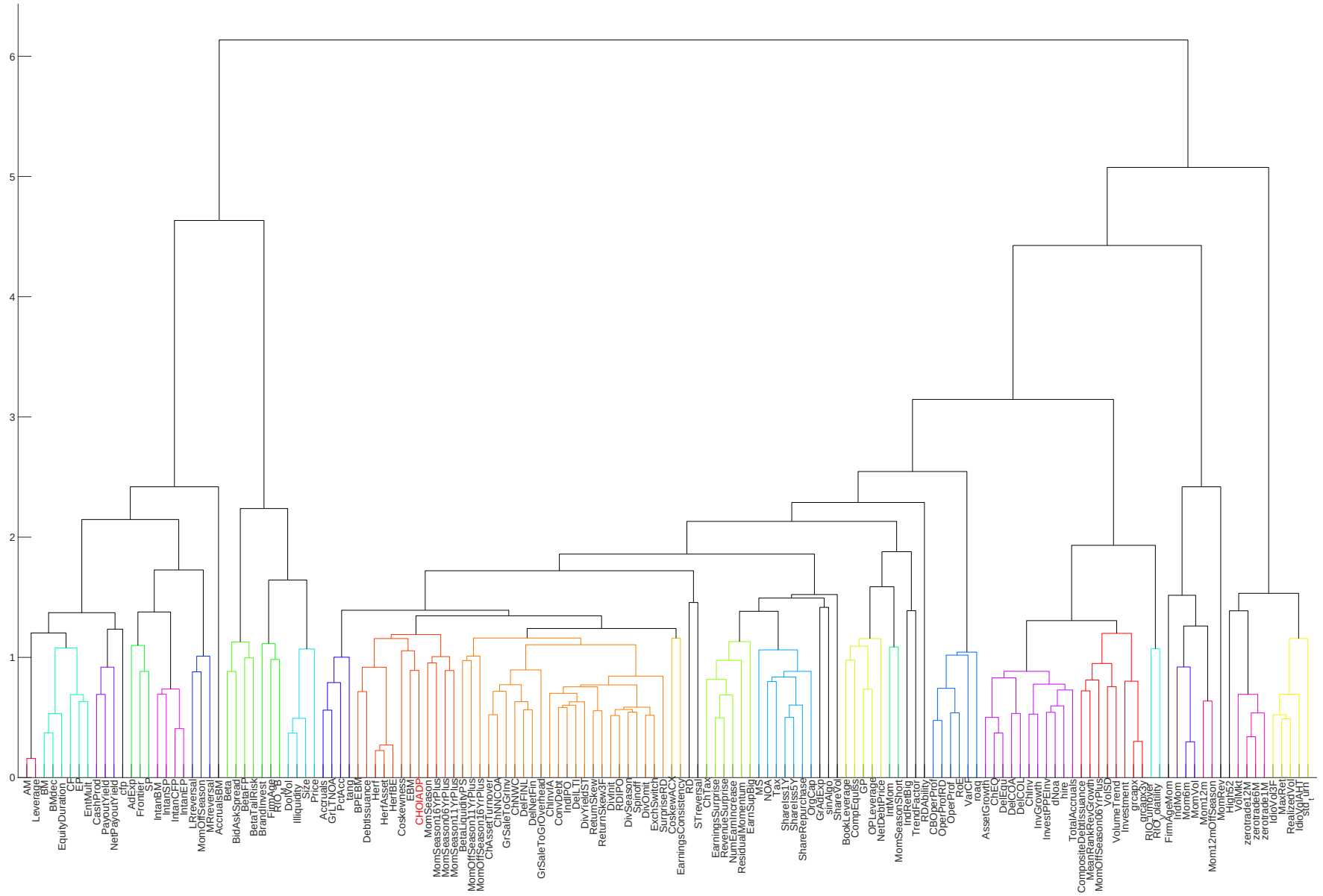


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

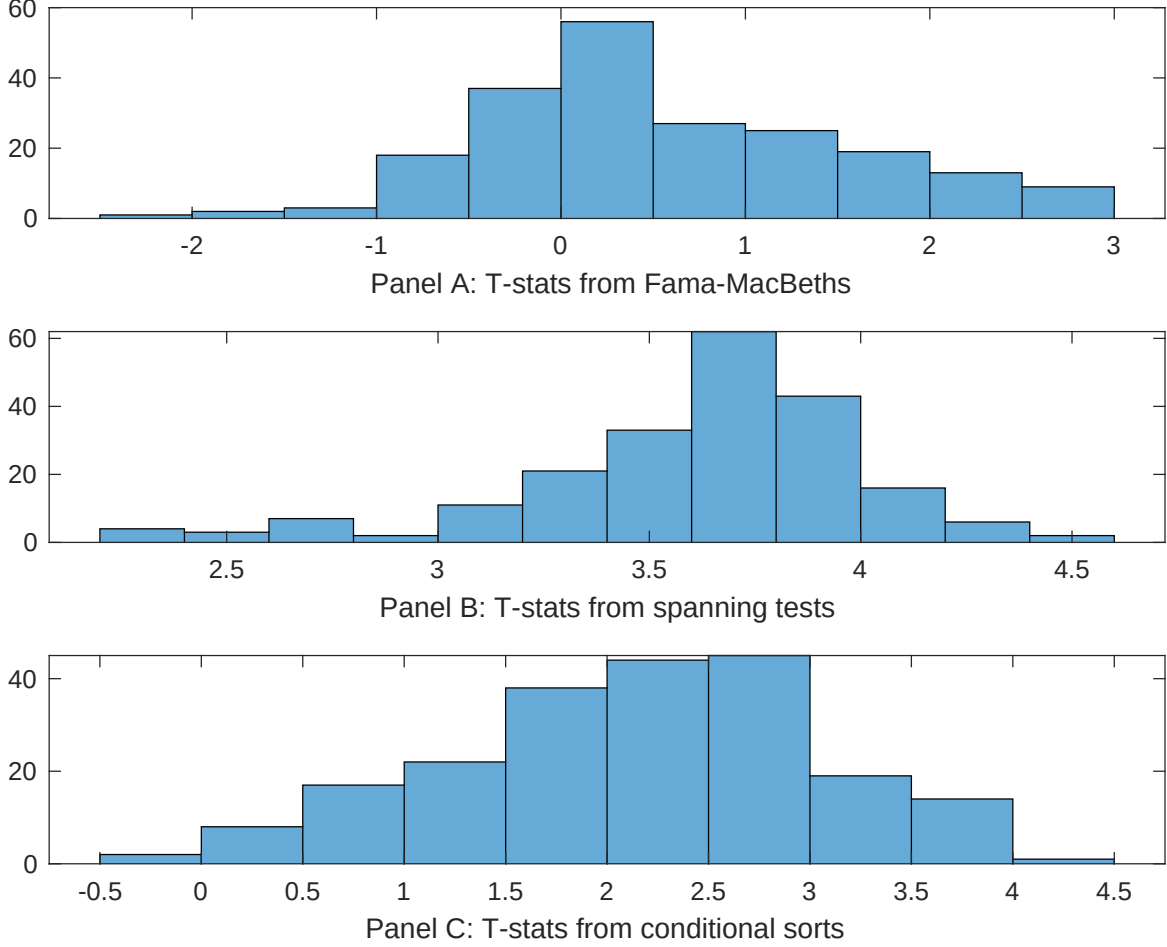


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of OLC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OLC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OLC} OLC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OLC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OLC. Stocks are finally grouped into five OLC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OLC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OLC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OLC} OLC_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are net income / book equity, Price, Idiosyncratic risk (3 factor), Idiosyncratic risk (AHT), Realized (Total) Volatility, Cash-flow to price variance. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.12 [4.95]	0.12 [4.28]	0.15 [8.03]	0.14 [7.51]	0.15 [8.73]	0.12 [5.48]	0.17 [8.79]
OLC	-0.58 [-0.08]	0.53 [0.08]	-0.49 [-0.82]	-0.45 [-0.84]	-0.53 [-0.91]	0.37 [0.52]	-0.31 [-0.58]
Anomaly 1	0.43 [0.03]						0.12 [0.15]
Anomaly 2		0.61 [1.28]					0.90 [3.21]
Anomaly 3			0.16 [3.58]				-0.12 [-1.05]
Anomaly 4				0.96 [1.71]			-0.20 [-0.45]
Anomaly 5					0.15 [3.55]		0.28 [2.53]
Anomaly 6						-0.94 [-0.82]	-0.13 [-2.70]
# months	648	648	643	643	643	643	643
$\bar{R}^2(\%)$	0	1	2	2	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the OLC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OLC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are net income / book equity, Price, Idiosyncratic risk (3 factor), Idiosyncratic risk (AHT), Realized (Total) Volatility, Cash-flow to price variance. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.35 [3.23]	0.35 [3.22]	0.33 [3.04]	0.32 [2.99]	0.33 [3.05]	0.33 [3.02]	0.34 [3.10]
Anomaly 1	-14.39 [-2.38]						-18.46 [-2.78]
Anomaly 2		-6.49 [-1.45]					-4.31 [-0.84]
Anomaly 3			0.06 [0.02]				-5.81 [-0.77]
Anomaly 4				5.20 [1.40]			11.03 [1.81]
Anomaly 5					0.90 [0.28]		-1.34 [-0.23]
Anomaly 6						2.42 [0.58]	3.04 [0.61]
mkt	14.29 [5.35]	17.40 [6.65]	16.45 [5.77]	18.44 [6.34]	16.86 [5.70]	17.14 [6.09]	16.90 [5.53]
smb	0.79 [0.19]	12.27 [2.03]	5.68 [1.15]	10.75 [2.06]	6.28 [1.43]	7.35 [1.54]	11.46 [1.79]
hml	7.44 [1.50]	9.91 [2.00]	8.85 [1.75]	7.37 [1.46]	8.57 [1.70]	9.92 [1.89]	7.74 [1.32]
rmw	2.15 [0.29]	-14.66 [-2.66]	-11.22 [-1.97]	-16.04 [-2.65]	-11.91 [-2.12]	-11.98 [-2.32]	-2.49 [-0.31]
cma	-2.65 [-0.36]	0.84 [0.11]	0.73 [0.10]	-1.31 [-0.18]	0.30 [0.04]	1.32 [0.18]	-3.82 [-0.48]
umd	-4.80 [-1.91]	-8.86 [-2.47]	-5.13 [-1.95]	-5.97 [-2.30]	-5.31 [-2.03]	-5.48 [-2.11]	-8.00 [-2.19]
# months	684	684	680	680	680	680	680
$\bar{R}^2(\%)$	12	11	11	11	11	11	12

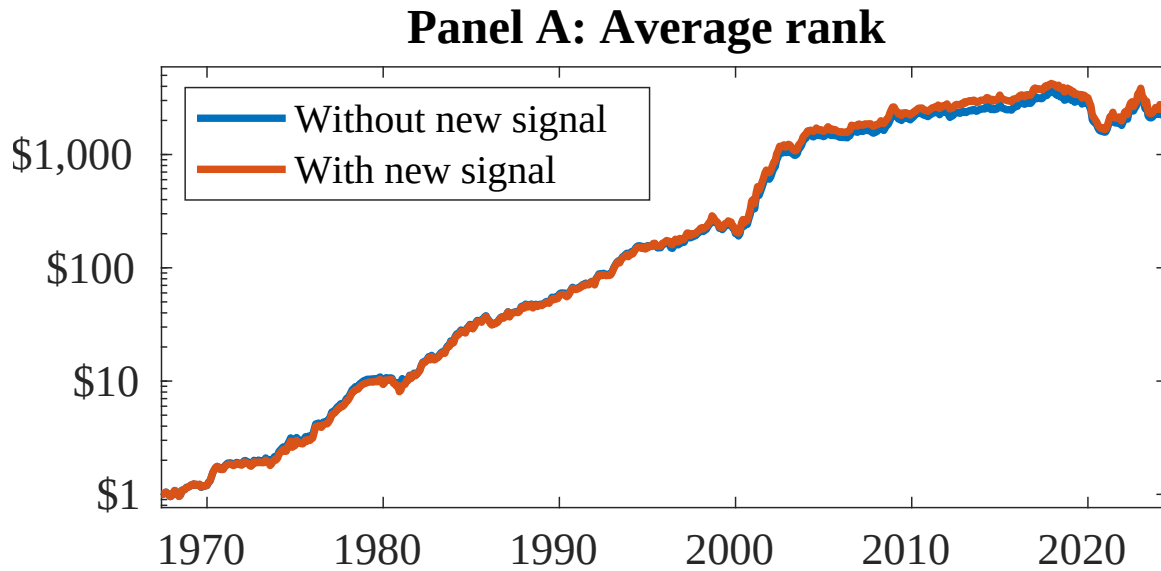


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as OLC. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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